MathMax Analytics and Fluency Scoring System: Detailed Explanation for New Joiners

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Introduction

This document provides a detailed explanation of the MathMax project, focusing on the analytics and fluency scoring systems. It is structured to help new team members understand how the system works and how data is collected, stored, and analyzed.

Part 1: Analytics System

1. Question Metadata Table

The Question Metadata Table stores all unique questions generated for students across different operations and difficulty levels. This ensures that we do not repeat questions unnecessarily and can track performance trends.

question_id	operation	domain	range	question_text	correct_answer	base_xp	tags	total_attempts	correct_count	incorrect_count	skip_count	avg_time	diffi
Q001	Addition	Single-digit	$1 \leq x \leq 5$	2+3	5	10	addition	20	18	2	0	1.8	1
Q002	Subtraction	Single-digit	$1 \leq x \leq 5$	5-2	3	10	subtraction	15	13	2	0	2.0	1
Q003	Multiplication	Single-digit	$1 \leq x \leq 5$	3*2	6	12	multiplication	12	11	1	0	2.2	2

Purpose: Helps detect difficult questions, track skipped or incorrect attempts, and feed adaptive learning recommendations.

2. Player Session Table (Sprint Tracking)

Each student interaction in a "sprint" session is recorded to track individual progress.

student_id	session_id	level_id	question_id	time_taken	result	xp_earned	fluency_points
S001	SPR001	add.1	Q001	1.8	Correct	10	10
S001	SPR001	add.1	Q002	2.0	Correct	10	10
S001	SPR001	multi.1	Q003	2.2	Correct	12	12

Purpose: Calculate session-level metrics, monitor fluency, and feed leaderboard data.

3. XP and Fluency Points

- Base XP is assigned per question based on difficulty.
- Fluency Points (FP) account for both accuracy and speed.
- Incorrect or skipped questions receive 0 FP.
- Leaderboards use cumulative FP to rank students.
- Adaptive difficulty adjusts future question recommendations.

Part 2: Scoring and Fluency Calculation System

1. Definitions

- Base XP (B_XP): Points per question based on difficulty.
- Expected Time (T_exp): Time considered standard for mental calculation.
- Actual Time (T_act): Time taken by the student to answer.
- Result (R): 1 if correct, 0 if incorrect, -1 if skipped.

2. Fluency Points Calculation

$$FP = \begin{cases} B\_XP \times \min\left(\frac{T\_exp}{T\_act}, 1.5\right), & \text{if correct} \\ 0, & \text{if incorrect or skipped} \end{cases}$$

Explanation: Faster correct answers yield higher FP; the cap prevents extreme bonuses. Skipped or incorrect questions are scored 0 FP but tracked for analytics.

Example

Level add.4, Base XP = 18, Expected Time = 4 sec

- Student answers in 3 sec $\Rightarrow$ FP $\approx$ 24
- Student answers in 5 sec $\Rightarrow$ FP = 14.4

3. Session Score Calculation

$$Total\ FP = \sum_{i=1}^{N} FP_i$$

$$Accuracy = \frac{\text{Correct Answers}}{\text{Total Attempted}} \times 100\%$$

4. Leaderboard XP Calculation

- Base XP scaled by difficulty multiplier.
- Fluency Points add bonus XP.
- Skipped questions = 0 FP but tracked.
- Cumulative XP sums FP across all sessions.

5. Fluency Score per Level

$$Level\ Fluency\ \% = \frac{Total\ FP\ earned}{Maximum\ FP\ possible} \times 100$$

Pass Threshold example: 70%

- LF% ¡70 → extra practice.
- LF% 70–90 → reinforce learning.
- LF% ¿90 → advance to next level.

6. Error and Skip Handling

- Incorrect answers receive 0 FP, tracked for identifying weak areas.
- Skipped questions receive 0 FP, but are included in review sessions.
- Both inform adaptive recommendations for students.

7. Summary Flow for New Joiners

1. Student attempts a question → record Actual Time (T_act) and Result.
2. Calculate Fluency Points (FP) using the formula.
3. Update session total FP.
4. Update Level Fluency %.
5. Update cumulative XP for leaderboard rankings.
6. Analyze skipped and incorrect questions to recommend targeted practice.